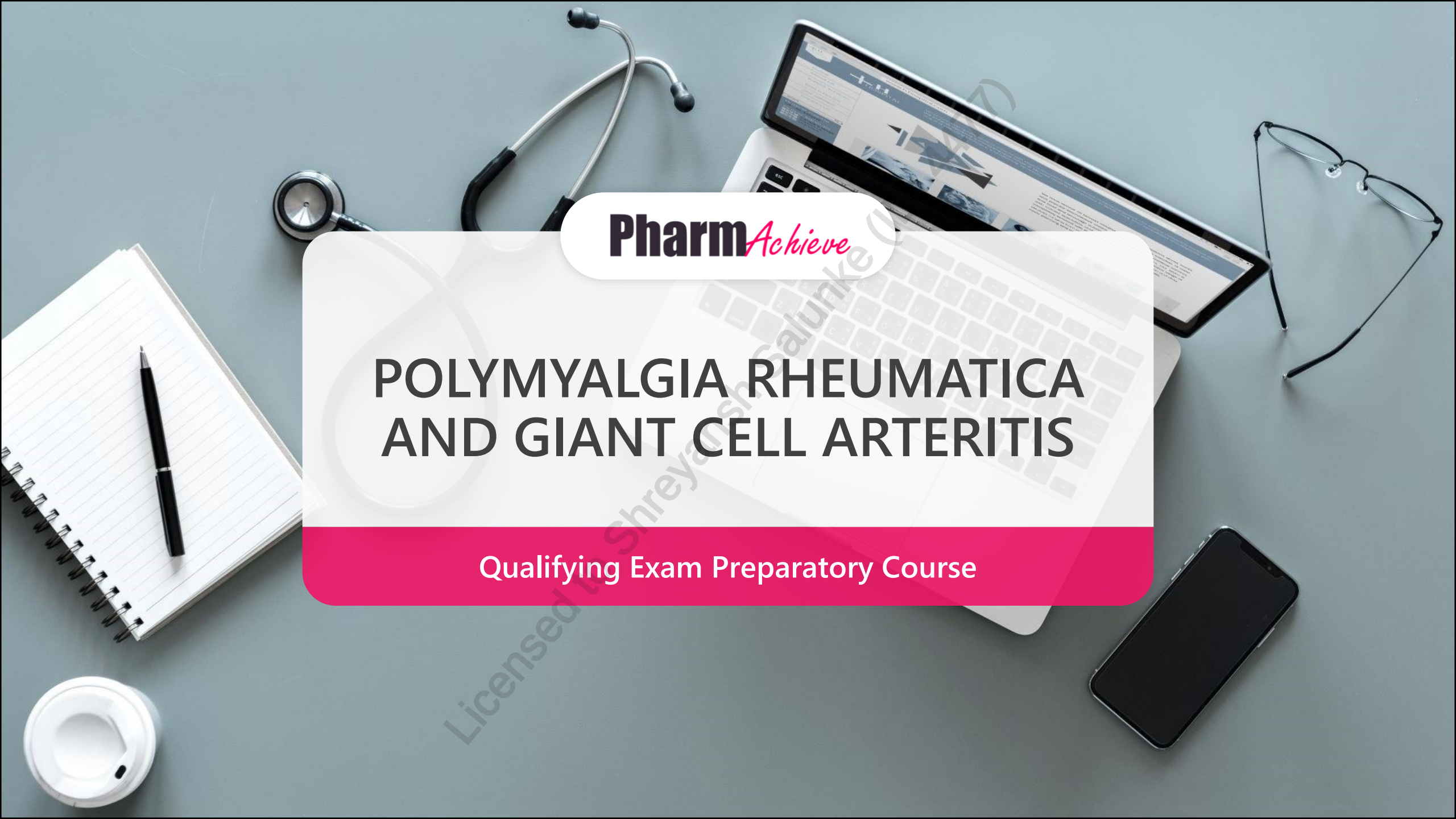


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop screen.

Pharm*Achieve*

POLYMYALGIA RHEUMATICA AND GIANT CELL ARTERITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

Anti-CCP Anticyclic Citrullinated Peptide Antibodies

CPK Creatine Phosphokinase

CRP C-Reactive Protein

EMG Electromyography

ESR Erythrocyte Sedimentation Rate

GCA Giant Cell Arteritis

IL Interleukin

MRI Magnetic Resonance Imaging

MSK Musculoskeletal

NSAID Non-Steroidal Anti-Inflammatory Drug

PMR Polymyalgia Rheumatica

PO Per Os (By Mouth)

RF Rheumatoid Factor

DEFINITIONS

- PMR and GCA are two separate but linked conditions which can occur together. PMR occurs more frequently than GCA
- They both occur in people ≥ 50 years old and are more common in women than men
- The percentage of patients with PMR who experience GCA at some point varies in reported series, approximating 10-21%. Approximately 50% of patients with GCA will develop PMR
- **Unknown** etiology

Polymyalgia Rheumatica (PMR)

- Inflammatory disorder linked to musculoskeletal aching and stiffness in the neck, shoulders and hips and thighs that follows a bilateral pattern
- Characterized by low-grade synovitis of the proximal joints

Giant Cell Arteritis (GCA)

- Causes chronic swelling of arteries (vasculitis) in mainly the scalp and head (does NOT affect the brain's blood vessels)
Affects cranial branches of the arteries coming from the aortic arch
- Granulomatous inflammation causes headaches, blurred vision, jaw and tongue pain, tenderness over the temples
- Sometimes referred to as "temporal arteritis"

PMR: CLINICAL PRESENTATION

History	Physical examination	Possible abnormal lab tests
• Proximal stiffness (neck, shoulders, buttocks and thighs) with functional limitations • Pain with acute to severe onset • Morning stiffness >45 minutes (hallmark symptom) • Symptoms worsen during the night and movement while sleeping causes severe pain that could wake the patient up • Systemic symptoms: low-grade fever, anorexia, fatigue, depression, and malaise	• Nonspecific findings such as a reduction in movement around the neck and shoulders • Small and large joint synovitis can occur but are unusual in areas distal to the ankle and wrist • Swelling with pitting edema on the back of the hands and feet • Proximal muscle tenderness • Muscle weakness is NOT a key symptom in patients with PMR	• Normocytic anemia • Elevated CRP, ESR • Thrombocytosis

GCA: CLINICAL PRESENTATION

History	Physical examination	Possible abnormal lab tests
• Headache and tenderness of the scalp to touch in temporal/occipital regions • Jaw claudication, may affect the tongue • Transient or permanent vision loss in one eye which could affect the second eye (rarely) • Fever, fatigue, and weight loss may occur	• Scalp tenderness with decreased (or absent) pulsation of the temporal artery • May see loss of upper limb pulses in patients with aortic arch branch involvement	• Normochromic anemia • Elevated CRP, ESR • Thrombocytosis • Temporal artery biopsy is done if GCA is clinically suspected and is not normally recommended in patients with isolated PMR • A normal biopsy does NOT exclude the diagnosis • Corticosteroid therapy should not be withheld while waiting for the biopsy results

DIFFERENTIAL DIAGNOSIS OF PMR

FYI

Diagnosis	Features differentiating conditions from PMR
Myositis	• Muscle weakness on physical examination • Elevated creatine phosphokinase (CPK) • Abnormal electromyography (EMG)/MRI, elevated muscle enzymes
Fibromyalgia	• Often seen in younger patients (<50 years old) • Widespread MSK pain, stiffness, aching, fatigue at many soft tissue sites • Symptoms have a longstanding duration • Normal ESR, CRP, and hematocrit
Rheumatoid arthritis	• Synovitis seen distal to wrist and ankle • Rheumatoid factor (RF) is seropositive, measurement of anti-CCP is essential • Low dose prednisone has no reported benefit • Radiographic erosions
Cancer	• Some cancers may present with similar symptoms to PMR • There is NO association between PMR and cancer

GOALS OF THERAPY

- ✓ Relieve symptoms
- ✓ Prevent permanent visual disturbances and vision loss
- ✓ Monitor drug therapy and reduce the risk of corticosteroid toxicity
- ✓ Restore the patient's daily functioning

PMR: DRUG THERAPY – CORTICOSTEROIDS

- Prednisone commonly used, 12.5-25 mg/day
- Symptom improvement within 24-72 hours, resolution within days
- If there is no significant improvement after 1 week of therapy, the patient should be re-assessed
- The minimum effective prednisone dose should be used, typically over at least 12 months
- Tapering <10 mg/day should be started once clinical and biochemical remission (normal ESR or CRP, or both) is achieved
- Clinicians should monitor patients closely for corticosteroid toxicity



NSAIDs are ineffective overall (they may only provide partial improvement of symptoms)

Sarilumab (Kevzara®), an IL-6 inhibitor, was recently approved for PMR in patients with inadequate response to corticosteroids or those who experienced a relapse during corticosteroid tapering



Other therapies can be used, but evidence is less robust, therefore will not be discussed, e.g. some evidence on the use of methotrexate as a steroid-sparing agent

DRUG THERAPY FOR GCA - CORTICOSTEROIDS

- Patients with GCA require **higher doses of corticosteroids**: 40-60 mg/day of prednisone (or equivalent) continued until symptoms and laboratory abnormalities resolve
- The risk of vision loss is reduced by an early diagnosis and prompt treatment
- **When a patient experiences vision loss, corticosteroids most likely will NOT reverse it**
- Symptoms usually respond quickly to corticosteroids (days to weeks)
- Most patients are treated for 1-2 years, but it is not uncommon to require a longer treatment duration
- Steroid-sparing agents may be considered for patients with GCA who are at an increased risk of corticosteroid-related adverse effects: **tocilizumab, upadacitinib, methotrexate**



Alternate-day dosing of prednisone is NOT recommended as it increases the chance of relapse

PREVENTION OF CORTICOSTEROID TOXICITY

- Corticosteroids are associated with significant side effects
 - The dose may require adjustment for disease severity, comorbid factors, fracture risk, patient concerns, and adverse reactions
 - Clinicians should give the **lowest possible dose** of corticosteroid required to control symptoms for the **shortest duration possible**
- Prednisone therapy with doses ≥ 7.5 mg daily for > 3 months has been linked with significant bone loss
 - Bisphosphonates prevent the bone loss associated with corticosteroids and should be administered when patients begin high dose corticosteroid therapy (≥ 30 mg/day and cumulative dose > 5 g in 1 year)

MONITORING PARAMETERS

Efficacy & Safety Endpoints		
Parameter	Degree of Change	Timeframe
Joint pain (PMR)	Improved	Day(s) to weeks
Visual acuity (GCA)	Improved	Days to weeks
ADR from medication therapy (various, depending on agent) Careful assessment of long-term corticosteroid therapy	Prevent	Duration of therapy
Recurrent symptoms	Prevent	Ongoing

THERAPEUTIC TIPS

Early diagnosis and treatment is important to prevent vision loss in patients with GCA

- Treatment should NOT be delayed while waiting for temporal artery biopsy results as biopsies can stay positive for up to 4 weeks after initiating corticosteroid therapy

Patients may present with a temporary increase in musculoskeletal symptoms after each reduction in corticosteroid dose

- These symptoms will subside over the following week

The ESR and CRP values usually parallel disease activity in patients with PMR and GCA

- These values can be used to *confirm* clinical suspicions of a disease flare but should NOT be used alone to make treatment decisions

Reduction in shoulder range of motion can occur in patients with PMR because of localized rotator cuff tendonitis or capsulitis (delay in diagnosis and drug therapy initiation increases the chances of occurrence)

- A local corticosteroid injection (e.g. methylprednisolone acetate) is usually helpful

CASE 1

OB is a 62-year-old female who has been experiencing frequent headaches for a few months. She presents to your family health clinic, and on examination the physician notes tenderness of the temporal artery with absent pulsation. She mentions that she experiences muscle pain and stiffness in her neck and shoulders, especially at night which has also affected her sleep. Her blood test indicates elevated CRP, normal CPK, thrombocytosis and anemia. The examination indicates no muscle weakness and a biopsy of the temporal artery is done. OB takes amlodipine 5 mg PO daily for her hypertension. She does not smoke and drinks alcohol occasionally.

1. Which of the following diagnoses can be ruled out based on OB's presentation and history?
 - a) GCA
 - b) Cancer
 - c) Myositis
 - d) PMR



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2. OB's biopsy confirms the diagnosis of GCA and PMR and she is prescribed prednisone 40 mg/day. Which of the following is NOT a risk factor for developing osteoporosis in this patient?
- a) Gender
 - b) Age
 - c) Hypertension
 - d) High-dose corticosteroid



CASE 1

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3. OB comes to fill her prescription at your pharmacy and is concerned about her new diagnosis. What advice do you NOT give OB to navigate her condition?
- a) Regular eye exam
 - b) Use of vitamin D supplement 400 units daily
 - c) Increasing her calcium intake to 1200 mg daily
 - d) Starting bisphosphonates



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CHANGE LOG

June 2025

- Formatting changes made throughout
- Duplicate information deleted
- Slides 14-16: Case study added

August 2025

- Formatting, grammar and spelling changes throughout
- Slide 3: Updated legend
- Slide 4: Updated epidemiology notes
- Slides 5, 6, 7: Updated wording
- Slide 9: Modified treatment duration, added tapering note
- Slide 10: Added point regarding symptomatic improvement and steroid-sparing agents
- Slide 11: Added dosage considerations point, changed $>$ to $\geq$ for prednisone dose associated with bone loss
- Slide 16: Updated units for vitamin D from 'm' to 'units'
- Slide 17: Updated references

November 2025

- No content changes made

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 9: Updated information and added a note on sarilumab use in PMR therapy